R13

Code No: 115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, May/June - 2019
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

used paper 25-05-2019 PM

PART - A

(25 Marks)

1.a)	Why lexical and syntax analyzers are separated?	[2]
b)	List the various error recovery strategies for a lexical analysis.	[3]
c)	Mention the types of LR parser.	[2]
d)	Define LR(0) items with examples.	[3]
e)	What are the benefits of intermediate code generation?	[2]
f)	Explain about hashing.	[3]
g)	What is a basic block?	[2]
h)	Discuss about common sub expression elimination.	[3]
i)	How do you calculate the cost of an instruction?	[2]
j)	List out the common issues in the design of code generator.	[3]

PART - B

(50 Marks)

2. Explain the various phases of a compiler in detail. Also write down the output for the following expression after each phase a: =b*cd. [10]

OR

3. What is FIRST and FOLLOW? Explain the steps to compute FIRST and FOLLOW with an example. [10]

4. Check whether the following grammar is SLR (1) or not. Explain your answer with Reasons. [10]
S→ L=R S→R L→*R L→id R→L.

OR

5. Consider the grammar.
E → E + T E → T T → T * F T → F F → (E) / id
Construct CLR parsing table for the above grammar. Give the moves of the CLR parser on id * id + id. [10]

6. What is a three address code? Mention its types. How would you implement the three address statements? Explain with examples. [10]

OR

7. Describe in detail the syntax directed translation of case statements. [10]

8. What are steps needed to compute the next use information? [10]

OR

9. Discuss about the following:
   a) Copy Propagation
   b) Dead code Elimination and
   c) Code motion. [10]

10. Write the code generation for the d:=(a-b)+(a-c)+(a-c). [10]

OR

11. Write a code generation algorithm. Explain about the descriptor and function getreg(). Give an example. [10]

---ooOoo---

used paper 25-05-2019 PM